DAILY PRACTICE CALENDAR

Grade 7 Math • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **California Common Core standards for Grade 7**. Students who want an extra challenge can add **Algebra 1 Competition Math**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Adding & Subtracting Integers
2	Aug 23–Aug 27	Adding & Subtracting Rational Numbers
3	Aug 30–Sep 3	Multiplying & Dividing Rational Numbers
4	Sep 6–Sep 10	Rational Number Word Problems
5	Sep 13–Sep 17	Unit Rates with Fractions
6	Sep 20–Sep 24	Proportional Relationships
7	Sep 27–Oct 1	The Constant of Proportionality
8	Oct 4–Oct 8	Percents: Tax, Tip & Markup
9	Oct 11–Oct 15	Percent Change, Interest & Error
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	Equivalent Expressions
12	Nov 1–Nov 5	Two-Step Equations
13	Nov 8–Nov 12	Equation Word Problems
14	Nov 15–Nov 19	Inequalities
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Inequality Word Problems
16	Dec 6–Dec 10	Scale Drawings
17	Dec 13–Dec 17	Angle Relationships
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Triangles
19	Jan 10–Jan 14	Circumference of Circles
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Area of Circles
22	Jan 31–Feb 4	Area of Composite Figures
23	Feb 7–Feb 11	Cross Sections of Solids
24	Feb 14–Feb 18	Surface Area of Prisms
25	Feb 21–Feb 25	Volume of Prisms
26	Feb 28–Mar 3	Surface Area & Volume Problems
27	Mar 6–Mar 10	Probability Basics
28	Mar 13–Mar 17	Compound Events
29	Mar 20–Mar 24	Simulations & Experimental Probability
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Sampling & Making Inferences
31	Apr 10–Apr 14	Comparing Two Populations
32	Apr 17–Apr 21	Year-End Review: Numbers & Proportions
33	Apr 24–Apr 28	Year-End Review & Practice Test
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027

Unit 1: Operations with Rational Numbers

Adding & Subtracting Integers

Core: Same signs: add and keep the sign. Different signs: subtract and take the sign of the larger absolute value.

Mon Aug 16

Adding integers

TRY Compute $-8 + 13$.

Tue Aug 17

Subtracting integers

TRY Compute $4 - (-9)$.

Wed Aug 18

The number line

TRY Start at -3 and move 7 units left. Where do you land?

Thu Aug 19

Real-world integers

TRY A submarine at -120 m rises 45 m. What is its new position?

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027

Unit 1: Operations with Rational Numbers

Adding & Subtracting Rational Numbers

Core: The integer rules work for fractions and decimals too.

Mon Aug 23

Fractions

TRY Compute $-\frac{2}{3} + \frac{1}{4}$.

Tue Aug 24

Decimals

TRY Compute $-3.5 - 2.75$.

Wed Aug 25

Mixed numbers

TRY Compute $-1\frac{1}{2} - (-3\frac{3}{4})$.

Thu Aug 26

Distance

TRY Find the distance between -4.5 and 2.25 on a number line.

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027

Unit 1: Operations with Rational Numbers

Multiplying & Dividing Rational Numbers

Core: An even number of negative factors gives a positive product; an odd number gives a negative one.

Mon Aug 30

Multiplying

TRY Compute $\left(-\frac{2}{5}\right) \times 15$.

Tue Aug 31

Dividing

TRY Compute $-\frac{3}{4} \div \left(-\frac{1}{2}\right)$.

Wed Sep 1

Sign rules

TRY Compute $(-1)(-2)(-3)(4)$.

Thu Sep 2

Order of operations

TRY Compute $-12 \div 3 + (-2)(-5)$.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027

Unit 1: Operations with Rational Numbers

Rational Number Word Problems

Core: Every rational number is a terminating or repeating decimal.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Fractions to decimals

TRY Write $\frac{5}{8}$ and $\frac{2}{3}$ as decimals.

Tue Sep 7

Terminating or repeating?

TRY Does the decimal for $\frac{7}{20}$ terminate or repeat?

Wed Sep 8

Multi-step problems

TRY It is $4°$F and the temperature drops $2.5°$F each hour for 6 hours. What is the final temperature?

Thu Sep 9

Bank balances

TRY A balance of $-\$42.50$ gets a \$60 deposit and an \$18.25 charge. What is the new balance?

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027

Unit 2: Proportional Relationships & Percents

Unit Rates with Fractions

Core: Divide to find the amount for 1 unit.

Mon Sep 13

Complex fractions

TRY Leo walks $\frac{1}{2}$ mile in $\frac{1}{4}$ hour. What is his speed in miles per hour?

Tue Sep 14

Recipes

TRY Half a batch uses $\frac{2}{3}$ cup of flour. How much flour does a full batch use?

Wed Sep 15

Work rates

TRY Kim paints $\frac{3}{4}$ of a wall in $\frac{1}{2}$ hour. How many walls per hour is that?

Thu Sep 16

Comparing rates

TRY Ana walks $2\frac{1}{2}$ mi in $\frac{3}{4}$ h. Ben walks 3 mi in 1 h. Who is faster?

Fri Sep 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027

Unit 2: Proportional Relationships & Percents

Proportional Relationships

Core: Proportional: every ratio $\frac{y}{x}$ is the same, and the graph is a line through the origin.

Mon Sep 20

Tables

TRY A table has x: 2, 4, 6 and y: 5, 10, 15. Is it proportional?

Tue Sep 21

Not proportional

TRY A taxi charges \$3 plus \$2 per mile. Is the cost proportional to the miles?

Wed Sep 22

Graphs

TRY A line goes through $(0, 0)$ and $(4, 6)$. Is it proportional? What is k?

Thu Sep 23

Solving proportions

TRY Solve $\frac{x}{12} = \frac{5}{8}$.

Fri Sep 24

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027

Unit 2: Proportional Relationships & Percents

The Constant of Proportionality

Core: $y = kx$, where k is the unit rate.

Mon Sep 27

Finding k

TRY In a proportional relationship, $y = 21$ when $x = 3$. Find k.

Tue Sep 28

Writing equations

TRY Pencils cost \$0.35 each. Write an equation for the cost c of p pencils.

Wed Sep 29

The point $(1, k)$

TRY A graph of cost vs. pounds passes through $(1, 4.5)$. What does that point mean?

Thu Sep 30

Using $y = kx$

TRY If $y = 2.4x$, find y when $x = 15$.

Fri Oct 1

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027

Unit 2: Proportional Relationships & Percents

Percents: Tax, Tip & Markup

Core: New total = original $\times$ (1 + rate).

Mon Oct 4

Sales tax

TRY An \$80 item has 7.5% sales tax. What is the total?

Tue Oct 5

Tips

TRY A \$36 meal gets a 20% tip. What is the total?

Wed Oct 6

Markup

TRY A store buys a lamp for \$25 and marks it up 40%. What is the price?

Thu Oct 7

Discount, then tax

TRY \$50 shoes are 30% off, then 8% tax is added. What is the final cost?

Fri Oct 8

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Proportional Relationships & Percents

Percent Change, Interest & Error — **Core:** Percent change = amount of change ÷ original amount.

Mon Oct 11
Percent increase
TRY A price goes from $40 to $50. What is the percent increase?

Tue Oct 12
Percent decrease
TRY A class shrinks from 80 to 68 students. What is the percent decrease?

Wed Oct 13
Simple interest
TRY Find the simple interest on $500 at 4% per year for 3 years.

Thu Oct 14
Percent error
TRY Maya estimated 45 jelly beans; there were 50. What is her percent error?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Proportional Relationships & Percents

Review & Checkpoint 1 — **Core:** Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: rational numbers
TRY Compute -2.4×1.5.

Tue Oct 19
Review: proportions
TRY Solve $\frac{3}{8} = \frac{x}{40}$.

Wed Oct 20
Review: percents
TRY A $60 item is marked up 25%. What is the new price?

Thu Oct 21
Review: unit rates
TRY $\frac{3}{4}$ lb of cheese costs $6. What is the price per pound?

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Expressions, Equations & Inequalities

Equivalent Expressions — **Core:** Distribute (watch the negatives!) and combine like terms.

Mon Oct 25
Distributing negatives
TRY Expand $-3(2x - 5)$.

Tue Oct 26
Combining like terms
TRY Simplify $5a - 3 - 2a + 8$.

Wed Oct 27
Fraction coefficients
TRY Simplify $\frac{1}{2}x + \frac{3}{4}x$.

Thu Oct 28
Factoring
TRY Factor $8x - 12$.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Expressions, Equations & Inequalities

Two-Step Equations — **Core:** Undo adding or subtracting first, then undo multiplying or dividing.

Mon Nov 1
The form $px + q = r$
TRY Solve $3x + 7 = 25$.

Tue Nov 2
Negative numbers
TRY Solve $-4x - 5 = 11$.

Wed Nov 3
The form $p(x + q) = r$
TRY Solve $2(x + 3) = 18$.

Thu Nov 4
Fraction coefficients
TRY Solve $\frac{2}{5}x - 1 = 7$.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Expressions, Equations & Inequalities

Equation Word Problems **Core:** Name the variable, write the equation, solve, then check in the story.

Mon Nov 8
Perimeter
TRY A rectangle has perimeter 54 and length 15. Find its width.

Tue Nov 9
Phone plans
TRY A plan costs $20 a month plus $0.10 per text. The bill is $32. How many texts were sent?

Wed Nov 10
Sharing a bill
TRY 4 friends split a bill equally, and each adds a $3 tip. Each pays $18. What was the bill?

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Consecutive integers
TRY Two consecutive integers add to 47. Find them.

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Expressions, Equations & Inequalities

Inequalities **Core:** Solve like an equation, but flip the sign when you multiply or divide by a negative.

Mon Nov 15
One-step
TRY Solve $x - 4 > 9$.

Tue Nov 16
Two-step
TRY Solve $3x + 2 \leq 20$.

Wed Nov 17
Flip the sign
TRY Solve $-2x + 1 < 9$.

Thu Nov 18
Graphing
TRY Describe the graph of $x \geq -2$ on a number line.

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Expressions, Equations & Inequalities

Inequality Word Problems **Core:** "At least" means $\geq$; "at most" means $\leq$.

Mon Nov 29
At most
TRY You have $50. Tickets cost $8 each, and parking is $6. What is the most tickets you can buy?

Tue Nov 30
At least
TRY Jo needs at least 80 points and has 52. Each question is worth 4. What is the fewest questions she must get right?

Wed Dec 1
Savings
TRY Ray has $120 and saves $15 a week. After how many weeks will he first have more than $300?

Thu Dec 2
Sensible answers
TRY Why must the answer to Monday's problem be a whole number?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Scale Drawings, Angles & Circles

Scale Drawings **Core:** Scale factor = drawing length ÷ actual length.

Mon Dec 6
Map scales
TRY On a map, 1 cm stands for 25 km. How far is 7.2 cm on the map?

Tue Dec 7
Blueprints
TRY On a blueprint, 1 in. stands for 4 ft. A room is 3.5 in. by 4 in. How big is the real room?

Wed Dec 8
Scale factor
TRY A model 6 cm long shows an object 120 cm long. What is the scale?

Thu Dec 9
Scaling area
TRY A drawing uses a scale of 1 : 3. A shape on it has area 10 cm^2. What is the actual area?

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Scale Drawings, Angles & Circles

Angle Relationships

Core: Complementary angles add to 90°, supplementary to 180°; vertical angles are equal.

Mon Dec 13
Complementary angles
TRY Find the complement of 37°.

Tue Dec 14
Supplementary angles
TRY Find the supplement of 112°.

Wed Dec 15
Vertical angles
TRY Vertical angles measure $(2x + 10)^\circ$ and 50°. Find x.

Thu Dec 16
Angle equations
TRY Two complementary angles measure x° and $(2x + 15)^\circ$. Find x.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Scale Drawings, Angles & Circles

Triangles

Core: Angles add to 180°; any two sides must add to more than the third.

Mon Jan 3
Angle sum
TRY Two angles of a triangle are 47° and 68°. Find the third.

Tue Jan 4
Can it be a triangle?
TRY Can 3, 5 and 9 be the side lengths of a triangle?

Wed Jan 5
Unique triangles?
TRY Do the angles 50°, 60°, 70° make exactly one triangle?

Thu Jan 6
Exterior angles
TRY An exterior angle of a triangle is 125°, and one remote interior angle is 70°. Find the other.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Scale Drawings, Angles & Circles

Circumference of Circles

Core: $C = \pi d = 2\pi r$.

Mon Jan 10
From the diameter
TRY A circle has diameter 10. Find its circumference in terms of π.

Tue Jan 11
Approximating π
TRY A circle has radius 7. Use $\pi \approx \frac{22}{7}$ to find its circumference.

Wed Jan 12
Finding the diameter
TRY A circle has circumference 31.4. Using $\pi \approx 3.14$, find its diameter.

Thu Jan 13
Wheels
TRY A wheel has diameter 0.5 m. How far does it roll in 100 turns?

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Scale Drawings, Angles & Circles

Mid-Year Review & Checkpoint

Core: Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: rational numbers
TRY Compute $-\frac{5}{6} + \frac{1}{3}$.

Tue Jan 18
Review: percents
TRY A $120 jacket is 15% off. What is the sale price?

Wed Jan 19
Review: equations
TRY Solve $5x - 9 = 26$.

Thu Jan 20
Review: angles
TRY An angle and its supplement differ by 40°. Find the smaller angle.

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 4: Scale Drawings, Angles & Circles

Area of Circles — **Core:** $A = \pi r^2$.

Mon Jan 24
Area
TRY Find the area of a circle with radius 6 in terms of π.

Tue Jan 25
From the diameter
TRY A circle has diameter 14. Use $\pi \approx \frac{22}{7}$ to find its area.

Wed Jan 26
Semicircles
TRY Find the area of a semicircle with diameter 10.

Thu Jan 27
From the circumference
TRY A circle has circumference 20π. Find its area.

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Area, Surface Area & Volume

Area of Composite Figures — **Core:** Add the pieces and subtract the holes.

Mon Jan 31
Rectangle + semicircle
TRY A semicircle sits on the 6-unit side of a 10 × 6 rectangle. Find the total area.

Tue Feb 1
Shaded regions
TRY A circle with diameter 8 fits inside a square with side 8. Find the area inside the square but outside the circle.

Wed Feb 2
Trapezoids
TRY Find the area of a trapezoid with bases 12 and 8 and height 5.

Thu Feb 3
Running tracks
TRY A 10 × 6 rectangle has a semicircle of radius 3 on each short end. Find the total area.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Area, Surface Area & Volume

Cross Sections of Solids — **Core:** Slicing a solid makes a flat shape.

Mon Feb 7
Slicing a cube
TRY A cube is sliced parallel to one face. What shape is the cross section?

Tue Feb 8
Slicing a prism
TRY A rectangular prism is sliced parallel to its base. What shape is the cross section?

Wed Feb 9
Slicing a pyramid
TRY A square pyramid is sliced parallel to its base. What shape is the cross section?

Thu Feb 10
Slicing a cylinder
TRY A cylinder is sliced straight down through its center. What shape is the cross section?

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Area, Surface Area & Volume

Surface Area of Prisms — **Core:** Surface area is the sum of the areas of all the faces.

Mon Feb 14
Rectangular prisms
TRY Find the surface area of a 5 × 4 × 3 box.

Tue Feb 15
Cubes
TRY Find the surface area of a cube with edge 2.5.

Wed Feb 16
Triangular prisms
TRY A prism is 10 long, and its bases are right triangles with sides 3, 4, 5. Find its surface area.

Thu Feb 17
Painting a box
TRY You paint every face of a 6 × 4 × 3 box except the 6 × 4 bottom. What area do you paint?

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Area, Surface Area & Volume

Volume of Prisms **Core:** $V = Bh$: base area times height.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Rectangular prisms

TRY Find the volume of an $8 \times 5 \times 2.5$ box.

Tue Feb 22

Triangular prisms

TRY A prism has a triangular base of area 15 and length 12. Find its volume.

Wed Feb 23

Missing height

TRY A prism has volume 210 and base area 35. Find its height.

Thu Feb 24

Composite solids

TRY A $4 \times 4 \times 2$ box and a $4 \times 2 \times 2$ box are glued together. Find the total volume.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: Area, Surface Area & Volume

Surface Area & Volume Problems **Core:** Wrapping means surface area; filling means volume.

Mon Feb 28

Which one?

TRY Does wrapping a gift box use surface area or volume?

Tue Feb 29

Filling a tank

TRY A tank is 50 cm $\times$ 30 cm $\times$ 40 cm. How many liters does it hold? (1 L $= 1{,}000$ cm^3)

Wed Mar 1

Doubling edges

TRY Every edge of a box is doubled. Its volume is multiplied by what?

Thu Mar 2

Cost problems

TRY Paint costs \$0.50 per ft^2. How much does it cost to paint all faces of a $3 \times 2 \times 2$ ft box?

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Probability & Statistics

Probability Basics **Core:** P(event) = favorable outcomes ÷ total outcomes, a number from 0 to 1.

Mon Mar 6

The probability scale

TRY An event has probability 0.95. Is it likely or unlikely?

Tue Mar 7

Simple events

TRY A bag has 3 red, 5 blue and 2 green marbles. Find P(blue).

Wed Mar 8

Complements

TRY For the same bag, find P(not green).

Thu Mar 9

Expected results

TRY A fair spinner has 4 equal colors. About how many reds do you expect in 200 spins?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Probability & Statistics

Compound Events **Core:** List the sample space with an organized list, a table or a tree.

Mon Mar 13

Sample spaces

TRY You flip a coin and roll a die. How many outcomes are possible?

Tue Mar 14

Two coins

TRY Two coins are flipped. Find P(both heads).

Wed Mar 15

Dice tables

TRY Two dice are rolled. Find P(the sum is 7).

Thu Mar 16

Tree diagrams

TRY You pick a shirt (red, blue or white) and pants (black or tan) at random. Find P(red shirt and black pants).

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Probability & Statistics

Simulations & Experimental Probability

Core: Experimental probability comes from data; more trials bring it closer to the theory.

Mon Mar 20
Experimental probability
TRY A coin landed heads 28 times in 50 flips. What is the experimental P(heads)?

Tue Mar 21
Theory vs. experiment
TRY What is the theoretical P(heads) for a fair coin?

Wed Mar 22
Predictions
TRY A player made 18 of 24 free throws. Predict how many she will make in 40 shots.

Thu Mar 23
Designing a simulation
TRY How could you use a number cube to simulate an event with probability $\frac{1}{3}$?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Probability & Statistics

Sampling & Making Inferences

Core: A random sample is the best way to learn about a whole population.

Mon Apr 3
Biased samples
TRY To learn the school's favorite sport, Ty asks only the basketball team. Is that sample random or biased?

Tue Apr 4
Making predictions
TRY In a random sample of 50 students, 12 walk to school. The school has 600 students. About how many walk?

Wed Apr 5
Many samples
TRY Three random samples have means 42, 45 and 39. What is a good estimate of the population mean?

Thu Apr 6
Population vs. sample
TRY 200 of a town's 3,000 voters are surveyed. What is the population?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Probability & Statistics

Comparing Two Populations

Core: Compare the centers, and use the spread to judge how big the difference is.

Mon Apr 10
Comparing means
TRY Class A has mean score 82 and Class B has mean score 76. What is the difference?

Tue Apr 11
Using MAD
TRY Both classes have MAD 3. How many MADs apart are the means?

Wed Apr 12
Box plots
TRY Two box plots have medians 20 and 28, and both IQRs are 8. How many IQRs apart are the medians?

Thu Apr 13
Drawing conclusions
TRY Two groups' means differ by only 0.2 MAD. Is that a meaningful difference?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Year-End Review

Year-End Review: Numbers & Proportions

Core: Mix the skills from Units 1–2 and show your work.

Mon Apr 17
Review: rational numbers
TRY Compute $\left(-\frac{3}{4}\right)\left(-\frac{8}{9}\right)$.

Tue Apr 18
Review: proportional relationships
TRY A proportional relationship $y = kx$ passes through $(5, 17.5)$. Find k.

Wed Apr 19
Review: interest
TRY Find the simple interest on \$800 at 5% per year for 2 years.

Thu Apr 20
Timed set: 10 in 20 min
TRY Compute $-18 \div (-3) - 4 \times 2$.

Fri Apr 21
Weekly Quiz
10 mixed review problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Year-End Review

Year-End Review & Practice Test

Core: Practice a full test, then learn from every error.

Mon Apr 24

Review: equations

TRY Solve $-2(x-4)=14$.

Tue Apr 25

Review: circles

TRY Find the area of a circle with radius 5.

Wed Apr 26

Review: probability

TRY A fair number cube is rolled. Find P(even).

Thu Apr 27

Practice Test

Full-length Grade 7 practice test covering the whole year.

Fri Apr 28

Error-log review

Redo every missed problem from the test; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Year-End Review

Final Checkpoint & Celebration

Core: Show what you have learned and get ready for Grade 8.

Mon May 1

Light review

TRY Find the circumference of a circle with diameter 9.

Tue May 2

Mixed warm-up set

TRY Solve $4x+3>15$.

Wed May 3

Final Checkpoint

Covers the whole year.

Thu May 4

Checkpoint review

Go over solutions together; students explain one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for Grade 8.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 5
Tue: 13
Wed: -10
Thu: -75 m

Week 2
Mon: $-\frac{5}{12}$
Tue: -6.25
Wed: $2\frac{1}{4}$
Thu: 6.75

Week 3
Mon: -6
Tue: $\frac{3}{2}$
Wed: -24
Thu: 6

Week 4
Mon: 0.625 and $0.\overline{6}$
Tue: Terminates (0.35)
Wed: $-11°$F
Thu: $-\$0.75$

Week 5
Mon: 2 mph
Tue: $1\frac{1}{3}$ cups
Wed: $1\frac{1}{2}$
Thu: Ana ($3\frac{1}{3}$ mph)

Week 6
Mon: Yes ($k = 2.5$)
Tue: No (it does not start at 0)
Wed: Yes; $k = 1.5$
Thu: $x = 7.5$

Week 7
Mon: 7
Tue: $c = 0.35p$
Wed: The cost is \$4.50 per pound
Thu: 36

Week 8
Mon: \$86
Tue: \$43.20
Wed: \$35
Thu: \$37.80

Week 9
Mon: 25%
Tue: 15%
Wed: \$60
Thu: 10%

Week 10
Mon: -3.6
Tue: $x = 15$
Wed: \$75
Thu: \$8

Week 11
Mon: $-6x + 15$
Tue: $3a + 5$
Wed: $\frac{5}{4}x$
Thu: $4(2x - 3)$

Week 12
Mon: $x = 6$
Tue: $x = -4$
Wed: $x = 6$
Thu: $x = 20$

Week 13
Mon: 12
Tue: 120
Wed: \$60
Thu: 23 and 24

Week 14
Mon: $x > 13$
Tue: $x \leq 6$
Wed: $x > -4$
Thu: Closed dot at -2, shaded to the right

Week 15
Mon: 5
Tue: 7
Wed: 13 weeks
Thu: You cannot buy part of a ticket

Week 16
Mon: 180 km
Tue: 14 ft by 16 ft
Wed: 1 : 20
Thu: 90 cm^2

Week 17
Mon: $53°$
Tue: $68°$
Wed: $x = 20$
Thu: $x = 25$

Week 18
Mon: $65°$
Tue: No ($3 + 5 < 9$)
Wed: No: many triangles of different sizes
Thu: $55°$

Week 19
Mon: 10π
Tue: 44
Wed: 10
Thu: $50\pi \approx 157$ m

Week 20
Mon: $-\frac{1}{2}$
Tue: \$102
Wed: $x = 7$
Thu: $70°$

Week 21
Mon: 36π
Tue: 154
Wed: 12.5π
Thu: 100π

Week 22
Mon: $60 + 4.5\pi$
Tue: $64 - 16\pi$
Wed: 50
Thu: $60 + 9\pi$

Week 23
Mon: A square
Tue: A rectangle the same as the base
Wed: A smaller square
Thu: A rectangle

Week 24
Mon: 94
Tue: 37.5
Wed: 132
Thu: 84

Week 25
Mon: 100
Tue: 180
Wed: 6
Thu: 48

Week 26
Mon: Surface area
Tue: 60 L
Wed: 8
Thu: \$16

Week 27
Mon: Likely
Tue: $\frac{1}{2}$
Wed: $\frac{4}{5}$
Thu: 50

Week 28
Mon: 12
Tue: $\frac{1}{4}$
Wed: $\frac{1}{6}$
Thu: $\frac{1}{6}$

Week 29
Mon: 0.56
Tue: $\frac{1}{2}$
Wed: 30
Thu: Let a roll of 1 or 2 mean the event happens

Week 30
Mon: Biased
Tue: 144
Wed: 42
Thu: All 3,000 voters

Week 31
Mon: 6 points
Tue: 2
Wed: 1
Thu: Not really: the data overlap a lot

Week 32
Mon: $\frac{2}{3}$
Tue: 3.5
Wed: \$80
Thu: -2

Week 33
Mon: $x = -3$
Tue: 25π
Wed: $\frac{1}{2}$

Week 34
Mon: 9π
Tue: $x > 3$

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